



The Secret Code Breakers

An Introduction to Regular Expressions (Regex)

Grade: 6

Session: 1

Subject: Computer Science / Data Literacy

1. The Library Mystery & What is Regex?

Imagine you are in a massive library searching for a specific book on a high shelf. You finally locate it, but the title on the spine is torn and partially missing: **T I L _**.

Could the full title be **TILE**? Or **TILL**? Or **TILT**?

In digital information systems, we frequently deal with "missing pieces" or variable patterns. Rather than searching for just one rigid word, computers use **Regular Expressions** (commonly shortened to **Regex**) to search, match, and extract data based on flexible patterns.

NORMAL SEARCH

Looks for one exact match.

Example: Looking for specifically "House No. 123".

REGEX SEARCH

Looks for any text matching a specified pattern.

Example: Looking for ANY address that starts with "1" and contains exactly 3 digits total.



Why is Regex a Data Detective's Superpower?

Regex enables computers to parse through millions of user input logs, email addresses, order IDs, and phone records in a fraction of a second to verify accuracy or extract key details!

2. Key Regex Building Blocks (Metacharacters)

A. The Dot (**.**) — The Wildcard

The dot symbol represents **any single character** — whether it is a letter, a digit, a punctuation mark, or even a space.

Pattern: **R . m**

Matches: **Ram** ('.' = 'a'), **Rim** ('.' = 'i'), **Rom** ('.' = 'o')

Does NOT Match: **Room** (contains two characters between 'R' and 'm')

B. The Plus Sign (**+**) — One or More

The plus sign specifies that the **immediately preceding character** must appear **1 or more times** in sequence.

Pattern: **go +**

Matches: **go** (one 'o'), **goo** (two 'o's), **gooo** (three 'o's)

Does NOT Match: **g** (zero 'o's)

C. The Asterisk (*) — Zero or More

The asterisk specifies that the preceding character can appear **0 or more times** (meaning it is optional or can repeat multiple times).

Pattern: **colou*r**

Matches: **color** (zero 'u's), **colour** (one 'u'), **colouur** (multiple 'u's)

D. Character Classes ([]) — Custom Sets

Square brackets define a set of allowed characters for a single position.

- **[0-9]** matches any single digit from 0 to 9.
- **[a-z]** matches any single lowercase letter from a to z.
- **[A-Z]** matches any single uppercase letter from A to Z.

3. Guided Practice & Real-World Contexts

Case Study: The Postman's PIN Code Helper

In India, postal PIN codes are exactly **6 digits**, where the first digit ranges from 1 to 9, followed by 5 digits ranging from 0 to 9.

Regex Pattern: **[1-9][0-9]{5}**

Testing Text: "Send the gift to 400001 Mumbai, then to 110001 Delhi. Don't send it to 12345 Bangalore because that code is too short!"

- 400001 — **Matched!** (Valid 6-digit PIN code)
- 110001 — **Matched!** (Valid 6-digit PIN code)
- 12345 — **Failed!** (Only 5 digits long)

4. Detective's Cheat Sheet

SYMBOL / SYNTAX	MEANING	EXAMPLE MATCH
. (Dot)	Any single character	c.t matches cat , cut , c1t
+ (Plus)	Preceding character 1 or more times	hi+ matches hi , hii , hiii

SYMBOL / SYNTAX	MEANING	EXAMPLE MATCH
* (Asterisk)	Preceding character 0 or more times	ap*le matches ale , apple , apple
[0-9]	Any single numerical digit	Agent[0-9] matches Agent7
[a-z]	Any single lowercase letter	[a-z]at matches bat , cat , rat



Student Practice Worksheet

Test Your Secret Code Breaking Skills! | Name: _____

Instructions: Read each question carefully and circle the best answer or write your response in the space provided.

Question 1 (Multiple Choice)

What text strings will the Regex pattern `R.m` match?

- [A] Only "Ram"
- [B] "Ram", "Rim", and "Rom"
- [C] "Room" and "Roam"
- [D] Any word starting with R and ending with m regardless of length

Question 2 (Multiple Choice)

Which Regex symbol indicates that the preceding character must appear **one or more times**?

- [A] Dot (`.`)
- [B] Asterisk (`*`)
- [C] Plus sign (`+`)
- [D] Square Brackets (`[ ]`)

Question 3 (Multiple Choice)

How many total digits does an Indian postal PIN code contain according to the standard pattern `[1-9][0-9]{5}` ?

- [A] 5 digits
- [B] 6 digits
- [C] 10 digits
- [D] 12 digits

Question 4 (Pattern Building)

Sometimes people misspell country names due to typos (for example typing "India" as "Indya"). Write a single Regex pattern using the wildcard dot (`.`) that will successfully match both **"India"** and **"Indya"**:

Your Pattern: `Ind` _____ `a`

Question 5 (Short Answer)

Explain the difference between the **+** (**Plus**) symbol and the ***** (**Asterisk**) symbol in Regular Expressions. Give a brief example of a word that **a*** would match, but **a+** would NOT match.

Question 6 (Application Task)

Suppose you are creating a validation system for student identification codes. Each ID must start with the letter 'S', followed by exactly 3 numerical digits (e.g., S101, S999). Write down a character class pattern that matches this format:

Pattern: **S** -----

Question 7 (Analysis)

Look at the pattern **b[aeiou]t**. Which of the following words will match this pattern? (Circle ALL that apply):

bat **bet** **boot** **bit** **boat** **but**